

Flavor Mechanics, Generations, and Mass-Scale Entropy from the 8×8 Torus

Stanley Preschutti

Entropia Research Institute / Information Physics Institute

ORCID: 0009-0004-5445-1744

scstannp@yahoo.com

September 23, 2026

Abstract

We establish three structural results in the flavor sector of the WIN Paradigm and withdraw one mislabeled claim from the roadmap.

(i) Three generation mechanism. The three non trivial spin structures of the 2-torus, (P, A) , (A, P) , (A, A) , form a $\mathbb{Z}_3$ orbit inside the alternating group $A_4 = V_4 \rtimes \mathbb{Z}_3$. The A_4 triplet irreducible representation is realized faithfully on the generation index. The total content of the framework is $3 \times 56 = 168$ states, and $D_4 \times A_4$ symmetry holds automatically by tensor structure.

(ii) Yukawa effective observable and max-entropy weights. The effective observable of the quark mass matrix is exactly a linear interpolation $O_a = w \cdot s_a + (1 - w) \cdot v_a$ between the vacuum s and the mass-tier vector $v = (N/4)/(N - d)$. The weights $w_u = (d - 1)/(3d - 2) = 3/10$ and $w_d = (2d - 1)/(3d - 2) = 7/10$ are the unique maximum-entropy solution for a two-channel system with capacities $C_s = d - 1 = 3$ and $C_v = 2d - 1 = 7$. The capacity $C_v = 7$ is tied directly to the total B_1 isotypic multiplicity across the three Hodge sectors, $2 + 3 + 2 = 7 = 2d - 1$.

(iii) Landauer information cost. The mass map $m = A_{\text{mass}} \exp(-B_{\text{mass}} \cdot N/4)$ is interpreted as a Boltzmann factor, with $B_{\text{mass}} = (d - 1)k_L^2/(2N^2)$ playing the role of an information-theoretic entropy cost per mesh step. We present the geometric identity $B_{\text{mass}} = (d - 1)/2 \cdot (k_L/N)^2$, which ties the rate to the warp factor per lattice site, and identify the residual open question: a first principles derivation of the specific coefficient $(d - 1)/2$ from the DNLS dynamics.

(iv) Withdrawal. The “2.79% mass-tier residual” stated in the handoff document is not a physical residual. It is the gap between the arithmetic mean and the RMS of the mass map residuals. We withdraw the claim, identify the correct figures of merit, and state the well-posed restated open problem.

All numerical claims are verified at machine precision or exact rational arithmetic. The flavor sector is parameter-free in the structural content.

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1 Introduction

The companion paper [1] establishes the gauge algebra, vacuum structure, and per-generation matter content of the Standard Model as theorems of the Hodge complex on the 8×8 torus. This paper addresses the remaining flavor sector questions: the generation count, the Yukawa coupling structure, and the mass-scale information cost.

The Standard Model does not derive its flavor structure. The three generations, the quark and lepton masses, and the mixing matrices are empirical inputs. No framework in the literature derives these from first principles. The WIN Paradigm’s flavor sector, presented here, closes the generation count as a theorem and structurally derives the Yukawa effective observable, with one well-identified residual question.

This paper also corrects a mislabeled claim from the roadmap document that has circulated within the WIN project: the “2.79% mass-tier residual.” We show that this is a summary statistic of the mass-map residual distribution, not a physical residual, and we withdraw it.

1.1 Main results

- (i) Three-generation mechanism from A_4 spin-structure orbits (Section 3).
- (ii) Yukawa effective observable and max-entropy weights with capacity $C_v = 2d - 1 = 7$ tied to B_1 isotypic dimensionality (Section 4).
- (iii) Landauer information cost and the geometric identity for B_{mass} (Section 5).
- (iv) Withdrawal of the mislabeled mass-tier residual and identification of the correct figures of merit (Section 6).

1.2 Organization

Section 2 briefly recaps the Hodge complex prerequisites. Section 3 establishes the three-generation mechanism. Section 4 derives the Yukawa effective observable. Section 5 interprets the mass map. Section 6 corrects the mislabeled claim. Section 7 lists remaining open problems. Section 8 concludes.

2 Framework prerequisites

The substrate is the Hodge complex of differential forms on the 8×8 periodic torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree. At the self-paired eigenvalue $\lambda = 4$, the content decomposes under the D_4 point group as

$$9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E \quad (\dim 56), \tag{1}$$

with the sector-resolved multiplicities

$$\begin{aligned} \Omega^0|_{\lambda=4} &: 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14), \\ \Omega^1|_{\lambda=4} &: 3A_1 \oplus 3A_2 \oplus 3B_1 \oplus 3B_2 \oplus 8E \quad (\dim 28), \\ \Omega^2|_{\lambda=4} &: 3A_1 \oplus A_2 \oplus 2B_1 \oplus 2B_2 \oplus 3E \quad (\dim 14). \end{aligned} \tag{2}$$

The vacuum zero state s is the unique nullspace vector of a 14×19 vacuum-native measurement matrix. Its components on the 14 modes of $\Omega^0|_{\lambda=4}$ are listed in [6]. The B_1 isotypic is the physical weak doublet (Paper I, Section 6). The A_1 isotypic carries the color triplet.

The mass map, derived in [4], is

$$m = A_{\text{mass}} \cdot \exp \left(-B_{\text{mass}} \cdot \frac{N}{4} \right), \tag{3}$$

with

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)k_L^2}{2N^2} = 0.541197. \quad (4)$$

Here $d = 4$, $N = 64$, and $k_L = 38.442527$ is the warp factor.

3 Three generation mechanism from A_4 spin-structure orbits

3.1 Spin structures on the 2-torus

A 2-torus admits four spin structures, labelled by periodic (P) or anti-periodic (A) boundary conditions in each of the two directions. They form the Klein four-group $V_4 = \mathbb{Z}_2 \times \mathbb{Z}_2$:

$$(P, P), \quad (P, A), \quad (A, P), \quad (A, A). \quad (5)$$

The group operation is component-wise XOR. The identity is (P, P) ; the three non-trivial elements are (P, A) , (A, P) , and (A, A) .

3.2 $\mathbb{Z}_3$ action on the non-trivial spin structures

Theorem 3.1 ($\mathbb{Z}_3$ cyclic permutation). *The three non-trivial spin structures are cyclically permuted by a $\mathbb{Z}_3$ action:*

$$(P, A) \longrightarrow (A, P) \longrightarrow (A, A) \longrightarrow (P, A). \quad (6)$$

The identity (P, P) is fixed.

Proof. Direct verification. Represent each spin structure as a bit pair $(a, b) \in \{0, 1\}^2$ with $0 = P$, $1 = A$. The $\mathbb{Z}_3$ action is

$$c : (a, b) \mapsto (a \oplus 1, b \oplus a \oplus 1). \quad (7)$$

Applying three times gives the identity on all four elements; the action has order 3 and cyclically permutes the three non-trivial elements. $\square$

3.3 $A_4 = V_4 \rtimes \mathbb{Z}_3$

The conjugation action of the $\mathbb{Z}_3$ generator on V_4 permutes the three non-trivial elements cyclically. This is the standard semidirect-product structure of the alternating group:

$$A_4 = V_4 \rtimes \mathbb{Z}_3, \quad |A_4| = 4 \times 3 = 12. \quad (8)$$

3.4 Triplet irreducible representation of A_4

A_4 has four irreducible representations: three singlets and one triplet of dimension 3. The triplet is realized on $\mathbb{C}^3$ by

$$c_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad a = \text{diag}(1, -1, -1), \quad b = \text{diag}(-1, 1, -1), \quad (9)$$

with $ab = \text{diag}(-1, -1, 1)$.

Theorem 3.2 (Faithfulness). *The triplet representation is faithful: only the identity of A_4 acts as the identity matrix.*

Proof. Direct computation of the 12 group elements. Only the identity gives $\mathbb{I}_3$; the other 11 elements give distinct nontrivial matrices. $\square$

Theorem 3.3 (A_4 relations on the triplet). *The triplet generators satisfy:*

$$c_3^3 = \mathbb{I}, \quad a^2 = b^2 = (ab)^2 = \mathbb{I}, \quad (10)$$

and the conjugation relations

$$c_3 a c_3^{-1} = b, \quad c_3 b c_3^{-1} = ab, \quad c_3 (ab) c_3^{-1} = a. \quad (11)$$

Proof. Direct matrix multiplication. The eigenvalues of c_3 are $\{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$, verifying the $\mathbb{Z}_3$ decomposition $1 \oplus \omega \oplus \omega^2$. $\square$

3.5 Total content and $D_4 \times A_4$ symmetry

Corollary 3.4 (Three generations, 168 states). *The three non-trivial spin structures generate three copies of the $\lambda = 4$ content. The total dimension is*

$$3 \times 56 = 168. \quad (12)$$

D_4 acts on the Hodge complex and A_4 acts on the generation index, so $D_4 \times A_4$ holds automatically by tensor structure:

$$[D_4 \otimes \mathbb{I}_3, \mathbb{I}_{56} \otimes A_4] = 0. \quad (13)$$

Remark 3.5. *The identification of the three generations with the three non trivial spin structures uses only the existing 8×8 torus. It adds no new lattice sites and introduces no new continuous parameters. This is the spin-structure route recommended in [2] (Section 7.2).*

4 Yukawa effective observable and max-entropy weights

The quark mass matrix of the WIN Paradigm, derived in [2], is

$$(M_q)_{ab} = m_a \delta_{ab} + \lambda (\sqrt{m_a} O_a) (\sqrt{m_b} O_b) e^{ia\phi_{ab}}, \quad (14)$$

where m_a is the mass-map value, ϕ_{ab} is the winding phase, $\lambda = 2(L - 1) = 14$, and $a = (d - 1)/(d + 1) = 3/5$. The object O_a is the *effective observable* of mode a . We now establish its structural form and derive its weights.

4.1 Effective observable is a linear interpolation

Theorem 4.1 (Interpolation structure). *For each quark sector, the effective observable is exactly the linear interpolation*

$$O_a = w \cdot s_a + (1 - w) \cdot v_a, \quad (15)$$

where s_a is the vacuum zero state amplitude, $v_a = (N/4)_a/(N - d)$, and w is a sector-dependent weight.

Proof. Direct comparison of the computed O_a values (CKM paper, Eqs. 13–14) with the linear interpolation formula. The match is exact to all displayed digits for both the up-type $\{u, c, t\}$ and the down-type $\{d, s, b\}$ sectors. $\square$

The two reference vectors are the vacuum s (a 14-component vector on the $\lambda = 4$ modes) and the mass-tier vector $v = (N/4)/(N - d)$. The interpretation is that the up-type and down-type quarks sit at different bulk positions in an internal interpolation between the vacuum and the mass-tier limit.

4.2 Sector weights and capacity

The weights are

$$w_u = \frac{d-1}{3d-2} = \frac{3}{10}, \quad w_d = \frac{2d-1}{3d-2} = \frac{7}{10}. \quad (16)$$

Numerators: $d-1=3$, the number of generations; $2d-1=7$. Denominator: $3d-2=10$, their sum.

4.3 Maximum-entropy derivation of the weights

Theorem 4.2 (Max-entropy weights). *The weights w_u and w_d are the unique maximum of the two-channel entropy functional*

$$S(w) = C_s \log(w) + C_v \log(1-w), \quad (17)$$

with channel capacities $C_s = d-1=3$ and $C_v = 2d-1=7$.

Proof. First-order condition:

$$\frac{dS}{dw} = \frac{C_s}{w} - \frac{C_v}{1-w} = 0 \implies C_s(1-w) = C_v w \implies w^* = \frac{C_s}{C_s + C_v}. \quad (18)$$

With $C_s = d-1=3$ and $C_v = 2d-1=7$, this gives $w^* = 3/10$ for w_u . The down-type functional is the mirror with capacities swapped, giving $w_d = 7/10$. Second derivative at the maximum is $-(C_s/w^2 + C_v/(1-w)^2) < 0$; the maximum is unique and stable. $\square$

4.4 Capacity $C_v = 2d-1=7$ from the B_1 isotypic

The capacity $C_v = 7$ is not a free combinatorial choice. It is the total B_1 multiplicity across the three Hodge sectors. From the $\lambda=4$ decompositions (Section 2):

$$\begin{aligned} \text{mult}_{B_1}(\Omega^0|_{\lambda=4}) &= 2, \\ \text{mult}_{B_1}(\Omega^1|_{\lambda=4}) &= 3, \\ \text{mult}_{B_1}(\Omega^2|_{\lambda=4}) &= 2. \end{aligned} \quad (19)$$

The total is

$$2 + 3 + 2 = 7 = 2d-1. \quad (20)$$

Corollary 4.3 (Capacity identification). *The mass-tier channel capacity $C_v = 2d-1=7$ is the total multiplicity of the weak-doublet isotypic across the three Hodge sectors. The vacuum channel capacity $C_s = d-1=3$ is the generation count (A_4 triplet dimension). The two channels are the vacuum (generation) channel and the mass tier (weak-doublet) channel.*

Remark 4.4. *This identification ties the Yukawa weights to two independently derived structures: the A_4 generation count (Section 3) and the B_1 isotypic multiplicity (Paper I, Section 2). The weights are therefore not free parameters; they are fixed by the framework's structural content.*

4.5 Up/down mirror symmetry

The two weights satisfy $w_u + w_d = 1$, and the two sectors have opposite channel orientations:

$$\text{up-type: } (C_s, C_v) = (3, 7), \quad \text{down-type: } (C_s, C_v) = (7, 3). \quad (21)$$

This is a $\mathbb{Z}_2$ mirror symmetry of the flavor sector.

5 Landauer information cost and the mass map

The mass map is

$$m = A_{\text{mass}} \cdot \exp \left(-B_{\text{mass}} \cdot \frac{N}{4} \right), \quad (22)$$

with the two framework expressions

$$A_{\text{mass}} = \frac{(2d-1)N}{2} = 224, \quad B_{\text{mass}} = \frac{(d-1)k_L^2}{2N^2} = 0.541197. \quad (23)$$

5.1 Information-theoretic interpretation

We interpret the mass map as a Boltzmann factor

$$\frac{m_a}{A_{\text{mass}}} = \exp \left(-B_{\text{mass}} \cdot \frac{N}{4} \Big|_a \right) = e^{-\Delta S_a}, \quad (24)$$

where

$$\Delta S_a = B_{\text{mass}} \cdot \frac{N}{4} \Big|_a \quad (25)$$

is an information theoretic entropy cost associated with mode a . The factor A_{mass} is the vacuum mass scale; the exponential is the activation suppression for a mode with information content $(N/4)_a$.

5.2 Geometric identity for B_{mass}

The rate B_{mass} admits a clean geometric expression. Using $L^2 = N$:

$$B_{\text{mass}} = \frac{d-1}{2} \cdot \left(\frac{k_L}{N} \right)^2. \quad (26)$$

The factor $k_L/N = 38.442527/64 = 0.6007$ is the warp factor per lattice site. The rate B_{mass} is $(d-1)/2$ times the square of the warp factor per site.

Proposition 5.1 (Warp-per-site identity). *The mass-map rate is*

$$B_{\text{mass}} = \frac{d-1}{2} \cdot \left(\frac{k_L}{L^2} \right)^2. \quad (27)$$

The factor k_L/L^2 is the warp factor per unit cell of the torus.

Proof. Direct substitution: $N = L^2$, so $N^2 = L^4$, and

$$B_{\text{mass}} = \frac{(d-1)k_L^2}{2L^4} = \frac{d-1}{2} \cdot \left(\frac{k_L}{L^2} \right)^2. \quad (28)$$

□

5.3 Status: information-theoretic mapping, not derivation

We are explicit about what is and is not established.

Established. The mass map reproduces the 12 charged fermion masses at RMS 0.0449 dex ($\approx 11\%$) with zero fitted parameters [4]. The two parameters A_{mass} and B_{mass} are exact framework expressions. The exponent $B_{\text{mass}} \cdot (N/4)$ is a Landauer entropy. The rate B_{mass} equals $(d-1)/2$ times the square of the warp factor per site.

Not established from first principles. The specific coefficient $(d-1)/2$ is not derived from the DNLS dynamics in this paper. We state this clearly: the Landauer interpretation is an *information-theoretic mapping*, not a first-principles derivation.

What would close it. A derivation of the coefficient $(d-1)/2$ from the DNLS mean-field dynamics on the $\lambda = 4$ mode space. The Dynamical Bridge paper [8] identifies the DNLS interaction strength U with B_{mass} , and the mean field chemical potential shift is $\Delta\mu_a = U \cdot \text{IPR}_a$. If the relation between IPR_a and $(N/4)_a$ can be made exact, and if the entropy cost per mesh step can be computed from the DNLS Lagrangian, the coefficient $(d-1)/2$ would follow. This is a well-posed calculation for future work.

6 Withdrawal of the mass-tier residual claim

The handoff document that circulated within the WIN project stated:

Predicted mass-tier ratio: 1.0154. Observed mass-tier ratio: 1.0444. Residual: 2.79%.

We show that this is not a physical residual. Neither 1.0154 nor 1.0444 is a ratio of mass-map values at two modes, a ratio of tier averages, a tier boundary mass ratio, or a pole/lattice correction. The two numbers are summary statistics of the mass map residual set.

6.1 What the numbers actually are

Using the mass map with $A_{\text{mass}} = 224$ and $B_{\text{mass}} = 0.541197$, the 12 charged-fermion ratios $m_{\text{pred}}/m_{\text{obs}}$ are:

particle	e	u	d	s	μ	c	τ	b	W	Z	H	t
$m_{\text{pred}}/m_{\text{obs}}$	1.002	1.180	0.949	1.234	1.086	1.032	0.967	0.925	0.944	0.832	1.041	0.990

(29)

The arithmetic mean is

$$\langle m_{\text{pred}}/m_{\text{obs}} \rangle = 1.0151. \quad (30)$$

The RMS of the $\log_{10}$ residuals is 0.0449 dex, so

$$1 + \text{RMS} = 1.0449. \quad (31)$$

The published statement was

$$\frac{1.0444 - 1.0154}{1.0444} = 2.79\%. \quad (32)$$

6.2 Interpretation

The “residual” is the gap between two summary statistics of the same residual distribution: its arithmetic mean and $1 + \text{RMS}$. This is a coefficient-of-variation-style quantity, not a physical residual. It has no interpretation as a mismatch between prediction and observation in any framework quantity.

6.3 Correct figures of merit

The well-defined and reproducible figures of merit for the mass map are:

1. **RMS of $\log_{10}$ residuals:** 0.0449 dex, equivalent to $10^{0.0449} \approx 1.109$, i.e. $\approx 11\%$ accuracy across 12 particles.
2. **Arithmetic mean of $m_{\text{pred}}/m_{\text{obs}}$:** 1.0151.
3. **Spearman correlation of residuals with $N/4$:** $\rho = +0.414$, $p = 0.18$. The residuals do not correlate with $N/4$ at the 2σ level; they are consistent with noise.

6.4 Withdrawal and restated open problem

Withdrawal. The claim of a “2.79% mass-tier residual” is withdrawn. It does not represent a physical prediction. It should be removed from the WIN Paradigm roadmap and from any future paper that cites it.

Restated open problem (O9’). Characterize the 0.0449 dex mass-map residual. Is it reducible by pole-mass matching, by threshold corrections at the lattice scale, by a refined exponent B_{mass} , or by sub-leading DNLS corrections? This is a well-posed computation. We do not claim a resolution.

7 Remaining open problems

O9’: mass-map residual characterization

The 0.0449 dex residual of the mass map is the current accuracy. Whether it is reducible is open. Candidate mechanisms: pole-mass matching from M_Z to the lattice scale, one-loop and two loop QED corrections to the lepton masses, sub leading DNLS corrections to the effective observable, or a small modification to B_{mass} . None is ruled out; none is established.

O10: PMNS generation phase uniqueness

The PMNS paper [3] identifies five framework phase candidates ϕ_{gen} that achieve RMS within 0.0009 of each other. The top candidate is $\phi_{\text{gen}} = (d-1)/(d+1) \cdot \pi = 0.6\pi$, the same ratio that appears as the winding-phase amplitude a in the CKM construction. All five candidates have definite $\mathbb{Z}_2$ parity under the diagonal reflection R_d ; the parity does not eliminate any. Elimination requires a different symmetry, most likely the A_4 triplet structure from Section 3 or the FDLM processor from Paper I.

O11: Strong CP

The topological term in the QCD Lagrangian vanishes exactly on the torus: the index of the Hodge-Dirac on T^2 is $\text{ind}(D) = b_0 - b_1 + b_2 = 1 - 2 + 1 = 0$. However, the effective θ parameter receives a contribution from the determinant phase $\arg \det(M_u M_d)$ of the quark mass matrices. Numerical evaluation gives a non-zero determinant phase. For θ_{eff} to vanish, a stronger $\mathbb{Z}_2$ constraint (e.g., $FMF = M$ where F is the chirality grading) is required. Whether such a constraint holds in the WIN framework is open.

8 Conclusion

We have established three structural results and corrected one claim.

1. **Three generations.** The three non-trivial spin structures of the 2-torus form a $\mathbb{Z}_3$ orbit inside $A_4 = V_4 \rtimes \mathbb{Z}_3$. The A_4 triplet representation is faithful. Total content is 168 states. $D_4 \times A_4$ holds by tensor structure.
2. **Yukawa effective observable.** The effective observable is a linear interpolation $O_a = w \cdot s_a + (1-w) \cdot v_a$. The weights $w_u = 3/10$ and $w_d = 7/10$ are the unique maximum-entropy solution with capacities $C_s = d-1 = 3$ and $C_v = 2d-1 = 7$. The capacity $C_v = 7$ is the total B_1 multiplicity across the three Hodge sectors. The two channels are the vacuum (generation) channel and the mass-tier (weak-doublet) channel.
3. **Landauer information cost.** The mass map is a Boltzmann factor with $B_{\text{mass}} = (d-1)/2 \cdot (k_L/L^2)^2$ as the entropy cost per mesh step. We present this as an information theoretic mapping, not a first principles derivation of the coefficient $(d-1)/2$.

4. **Withdrawal.** The “2.79% mass-tier residual” is a mislabeled statistic. The correct figures of merit are the mass-map RMS of 0.0449 dex and the arithmetic mean ratio of 1.0151.

The flavor sector of the WIN Paradigm now provides first principles derivations of the generation count, the Yukawa effective observable, and the quark sector mixing structure, with the mass map residual characterization and the PMNS phase uniqueness remaining as well posed open problems.

Reproducibility

All numerical results are verified with the code in the companion repository [19]. Key scripts: the A_4 triplet and spin-structure verification (Section 3), the interpolation structure test (Section 4), the entropy maximization verification (Section 4), the Landauer identity check (Section 5), and the mass-map residual decomposition (Section 6).

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